EMLM-Net: An Extended Multilinear Mixing Model-Inspired Dual-Stream Network for Unsupervised Nonlinear Hyperspectral Unmixing

Minglei Li, Bin Yang, *Member, IEEE*, and Bin Wang, *Senior Member, IEEE*

***Abstract*— To mitigate the impact of mixed pixels in hyperspectral images (HSIs), substantial progress has been made in both model- and deep-learning-based unmixing methods. However, the issues, such as complex computational processes and limited interpretability, hinder the improvement of their unmixing performance. Particularly, unsupervised nonlinear hyperspectral unmixing (HU) remains a great challenge. In this article, we propose an extended multilinear mixing (EMLM) model-inspired dual-stream network for unsupervised nonlinear HU. First, the alternating direction method of multipliers (ADMM) algorithm for the EMLM-based unmixing problem is unfolded to construct an encoder network. Subsequently, it is connected to a decoder network derived from the EMLM, creating an autoencoder (AE)-like network architecture. Second, the original HSIs and superpixel-averaging-based coarse HSIs are input into two network branches with identical architectures, respectively, to build a novel weight-sharing dual-stream network. Furthermore, the estimates of abundances and nonlinear parameters obtained from the two branches are utilized to formulate local spatial similarity regularizers, enhancing the network's loss function and effectively improving unmixing accuracy. Finally, experiments conducted on the laboratory-created dataset and real-world datasets validate that the proposed method exhibits superior unmixing performance compared to the state-of-the-art methods. In addition, our code is available at: https://github.com/I3ab/EMLM-Net.**

***Index Terms*— Autoencoder (AE) network, deep unfolding, hyperspectral images (HSIs), unsupervised nonlinear unmixing.**

I. Introduction

HYPERSPECTRAL images (HSIs) combine imaging spectroscopy with remote sensing (RS) techniques, providing detailed spectral information across contiguous bands from visible to near-infrared wavelengths. While offering spatial distributions of ground objects, HSIs are often affected by mixed pixels due to their limited spatial resolution and the diversity of natural ground surfaces, compromising application accuracy. To address this issue, hyperspectral unmixing (HU) is employed to decompose the mixed pixels to estimate pure material spectra (endmembers) and their corresponding proportions (abundances) [1].

The linear mixture model (LMM) assumes that light only interacts with one material in scenarios before reaching the sensor, i.e., observed spectra can be represented as linear combinations of endmembers. Owing to its straightforward mathematical form and physical interpretability, the LMM has found extensive use in unmixing [2]. The classical fully constrained least squares (FCLS) [3] method is commonly used for abundance estimation using endmembers extracted by methods like N-FINDR [4] and vertex component analysis (VCA) [5]. Nevertheless, in real scenarios, photon interactions between materials can cause nonlinear mixing effects.

In recent years, numerous nonlinear mixture models (NMMs) have emerged [6]. Drawing from the fundamental principles of radiative transfer theory, the Hapke model [7] has been formulated to describe the photon interactions of intimately mixed particles. However, the difficulty of acquiring this model's physical parameters limits its practical applicability. To address the second-order photon interactions, bilinear mixture models (BMMs) have been developed by incorporating additional bilinear interaction terms into the linearly combined spectra. Typical models include the Fan model [8], the generalized bilinear model (GBM) [9], and the polynomial postnonlinear mixing (PNMM) model [10]. Due to the simplicity of explaining nonlinearity in the BMMs, a series of unmixing methods have been proposed. For instance, methods based on the nonnegative matrix factorization (NMF) [11], [12] and those accounting for collinearity effects have been explored [13], [14]. Nevertheless, the influence of higher order scattering effects in real scenarios should not be underestimated. The multilinear mixing (MLM) model [15] considers infinite-order photon interactions by introducing probabilities for further photon interactions. Some existing researches have combined the block coordinate descent strategy [16], the manifold learning [17], and robustness metrics [18] with the MLM. However, these models implicitly assume that all spectral bands share the same mixing patterns, which might not accurately reflect real-world situations.

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Indeed, each band of electromagnetic waves interacts with ground covers in distinct ways, leading to varying levels of multiple scattering across different bands within the same pixel [19], [20], [21]. To explain wavelength-dependent nonlinear effects, an extended multilinear mixing (EMLM) model is derived by vectorizing the nonlinear parameters in the MLM. A bandwise nonlinear spectral unmixing (BNLSU) method [22] is proposed using the EMLM, which leverages the sparsity of abundances and the smoothness of nonlinear parameters to construct regularizers and employs the alternating direction method of multipliers (ADMM) [23] to solve the constrained optimization problem. To enhance the robustness of the EMLM-based unmixing method, a spectral–spatial reweighted robust nonlinear unmixing algorithm is introduced in [24]. However, these methods perform supervised unmixing, and the accuracy of abundance estimation relies on pre-extracted endmembers. A blind HU model called sparsity-enhanced convolutional decomposition is proposed in [25], which jointly captures the spatial–spectral information in a tensor-based fashion. However, due to the complex computational processes involved in the model-based constraint optimization problems, their applicability for real-time unmixing is limited.

In recent years, the rapid development of deep learning and artificial intelligence models has led to their growing applications in RS. In [26], a benchmark dataset is introduced for cross-city semantic segmentation using high-resolution domain adaptation networks. An interpretable deep unfolding network [27] is proposed for hyperspectral anomaly detection. Spectral GPT [28], a universal RS foundation model, is designed for spectral RS image processing using a 3-D generative pretrained transformer (GPT). Tackling pixel-level RS image classification tasks, a multimodal deep learning framework [29] is formulated. Additionally, an unsupervised 3-D point cloud semantic segmentation network, SegTrans [30], is implemented via transfer learning. To address the four degrees of freedom correspondence-based point set registration problem, a fast and deterministic method is presented in [31]. Rethinking HSI classification from a sequential perspective with transformers, a backbone network named SpectralFormer [32] is proposed.

Due to the powerful learning capabilities of deep neural networks, learning-based unmixing methods are gaining increasing attention [33], [34]. Different types of deep networks for HU have been proposed, with autoencoder (AE) networks being the primary representative. The core objective of AEs is to minimize the discrepancy between the input data and the reconstructed data, aiming to learn the low-dimensional representation of the former. In unmixing tasks, the encoder part of the AE maps the input data into hidden abundance estimates, while the decoder transforms the abundances back to the reconstructed data. Usually, the weights of the decoder can be regarded as the endmember spectra. A majority of deep-learning-based unmixing methods are based on the LMM. uDAS [35] introduces an untied AE network that applies denoising and sparsity constraints to the decoder and encoder, respectively. Gao et al. [36] employed two cascaded AEs in an end-to-end manner to tackle cycle-consistency constraints. In [37], a two-stream network is proposed to capture spectral and spatial information. Notably, these methods are tailored for solving linear unmixing problems that can be readily handled by classic methods. The potential of deep neural networks in addressing nonlinear problems has not been fully explored.

The intricate mathematical formulations and computations pose challenges for the application of deep neural networks in nonlinear unmixing. Nonetheless, some related exploratory works have been developed in recent years. In [38], a general deep learning framework called endmember-guided unmixing network (EGU-Net) is proposed for self-supervised nonlinear HU. Deep multitask bilinear unmixing (DMBU) [39] presents an unsupervised unmixing method based on the hierarchical BMM. In [40], an unsupervised nonlinear unmixing method is proposed, using the Hapke model and convolutional neural networks (HapkeCNN). A general nonlinear spectral mixture model is proposed by incorporating the nonlinear mixing component into the linear mixing term. To leverage the intrinsic characteristics of hyperspectral data, networks, such as the long short-term memory (LSTM) network [41] and the 3-D convolutional neural (3D-CNN) network [42], are adopted to derive nonlinear unmixing methods. However, the design of network architectures, especially that of encoder networks, lacks principled approaches for guidance. The lack of interpretability also makes it challenging to incorporate task-relevant priors into the design of network architectures.

The deep unfolding technique has emerged as a promising approach for designing interpretable network structures [43]. When applied to unmixing tasks, it provides guidance for designing network architectures, facilitates the incorporation of task-relevant priors into the network, and enables efficient real-time unmixing through end-to-end learning. Developing deep unfolding-based unmixing methods would be beneficial for addressing the issues in model- and deep learning-based methods. Related research indicates that networks derived from the unfolding of iterative algorithms often outperform their original iterative counterparts. The differences arise from the adopted iterative solvers, leading to variations in constructed unfolding network structures. Specifically, in [44], with the iterative shrinkage thresholding algorithm (ISTA) being unfolded, two model-inspired neural networks are designed. Besides, an interpretable sparsity-constrained NMF network [45] is introduced to alternately estimate endmembers and abundances. The ADMM algorithm for the constrained sparse regression problem is unfolded in [46], yielding two networks for supervised or unsupervised unmixing. However, these unfolding-based unmixing methods have some limitations. First, they are only applicable to simple linear unmixing, ignoring complex nonlinear mixing effects in real scenarios. Second, these methods perform pixelwise unmixing, lacking the utilization of the spatial information in HSIs.

Several studies have explored the integration of HSIs' spatial information into HU via superpixel segmentation. Generated based on the simple linear iterative clustering (SLIC) method [47], superpixels with irregular shapes represent local spatial neighborhoods. In a blind unmixing method named spatial group sparsity regularized NMF [48], superpixels act

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as a proxy for adaptively representing spatial groups. To fully exploit superpixel-based spatial similarity, an unsupervised nonlinear HU method introduces a coarse-to-fine scheme [49]. Another unsupervised HU approach, utilizing a two-stream AE network [50], takes superpixel segmentation as preprocessing to extract endmember bundles from HSIs. Leveraging superpixels to capture the local spatial similarity of HSIs proves to be an effective strategy for improving unmixing accuracy.

A. Motivations

The EMLM considers infinite-order photon scatterings and explains wavelength-dependent nonlinear mixing effects, making it a promising model for nonlinear unmixing. However, current studies on the EMLM are confined to numerical model optimization methods, which exhibit complex computational processes and limited unmixing performance. Inheriting the powerful learning capabilities of deep neural networks, deep-learning-based unmixing methods achieve high accuracy and efficiency in unmixing but suffer from poor physical interpretability. Combining the advantages of both the model- and deep-learning-based methods, various deep unfolding-based unmixing methods have been developed. However, these methods delve only into simple linear unmixing and neglect the prior knowledge of the local spatial similarity in HSIs.

B. Contributions

To design networks using deep unfolding techniques for unsupervised nonlinear HU and to effectively integrating the spatial information of HSIs into networks, we propose an EMLM-inspired dual-stream network, referred to as EMLM-Net. The contributions are summarized as follows:

1) For the first time, the deep unfolding technique is applied for nonlinear HU. A network, constructed by unfolding the ADMM algorithm for the EMLM-based unmixing problem, serves as an encoder, followed by a decoder derived from the EMLM, creating an AE-like network architecture. The network with such an architecture exhibits strong physical interpretability and supports unsupervised training, making it well-suited for solving the unsupervised nonlinear HU problem.
2) To fully exploit the spatial information of HSIs, a novel weight-sharing dual-stream network is proposed. Both branch networks employ an identical AE-like architecture and take the original HSIs and coarse HSIs as inputs, respectively. Their outputs, consisting of the estimates of abundances and nonlinear parameters, are utilized to construct the local spatial similarity regularizers for the loss function, leading the network to achieve accurate unmixing results.
3) Ablation experiments are conducted to illustrate the effectiveness of the proposed innovations in improving unmixing accuracy. Furthermore, the experimental results on different types of hyperspectral datasets demonstrate that the proposed method exhibits high unmixing performance, robustness to noise, strong physical interpretability, and good scalability compared to the state-of-the-art (SOTA) unmixing methods.

The remainder of this article is organized as follows. Section II briefly reviews the spectral mixture models and the EMLM-based constrained optimization problem. Next, the detailed derivation of the proposed method is provided in Section III. In Section IV, the experiments on different types of hyperspectral datasets are conducted. Finally, Section V gives the conclusion of this article.

II. Related Works

A. Spectral Mixture Models

1) LMM: Under the assumption of the LMM, for an HSI matrix $\boldsymbol{X} \in \mathbb{R}^{N\times M}$ containing N bands and M pixels, each pixel $\boldsymbol{x}_j \in \mathbb{R}^N$ $(j = 1, \ldots, M)$ can be represented as a linear combination of endmembers based on their abundances. Generally, the abundances should satisfy two constraints: the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC). The LMM can be described as

$$\boldsymbol{x}_j = \sum_{i=1}^{R} \boldsymbol{e}_i a_{i,j} + \boldsymbol{\varepsilon}_j, \quad \text{s.t. } a_{i,j} \geq 0, \ \sum_{i=1}^{R} a_{i,j} = 1 \tag{1}$$

where $\boldsymbol{e}_i \in \mathbb{R}^N$ is the ith endmember in the endmember matrix $\boldsymbol{E} \in \mathbb{R}^{N\times R}$, R is the number of endmembers, $a_{i,j}$ is the ith endmember's abundance in $\boldsymbol{x}_j$, and $\boldsymbol{\varepsilon}_j \in \mathbb{R}^N$ is the noise.

2) EMLM: By vectorizing the pixel-dependent nonlinear parameters of the MLM [15], the EMLM [22] is derived, which not only accounts for infinite-order photon scatterings but also captures the nonlinear intensity variations across different spectral bands. Specifically, the physical assumptions of the EMLM are as follows. First, before reaching the sensor, light interacts with at least one material. Then, for the jth pixel, the probability of light undergoing further interactions with materials, varying across different spectral bands, is denoted as $\boldsymbol{p}_j \in \mathbb{R}^N$, and the probability is proportional to the abundance corresponding to the material. Finally, the intensity of light scattered by a material is correlated with the material's reflectance. Therefore, the EMLM is formulated as

$$\begin{aligned}\boldsymbol{x}_j =& \left(\mathbf{1}_N - \boldsymbol{p}_j\right) \odot \sum_{i=1}^{R} \boldsymbol{e}_i a_{i,j} \\ &+ \left(\mathbf{1}_N - \boldsymbol{p}_j\right) \odot \boldsymbol{p}_j \odot \sum_{i,k=1}^{R} (\boldsymbol{e}_i \odot \boldsymbol{e}_k) a_{i,j} a_{k,j} \\ &+ \left(\mathbf{1}_N - \boldsymbol{p}_j\right) \odot \boldsymbol{p}_j^2 \\ &\odot \sum_{i,k,l=1}^{R} (\boldsymbol{e}_i \odot \boldsymbol{e}_k \odot \boldsymbol{e}_l) a_{i,j} a_{k,j} a_{l,j} + , \ldots, + \boldsymbol{\varepsilon}_j \\ =& \left(\mathbf{1}_N - \boldsymbol{p}_j\right) \odot \left(\boldsymbol{E}\boldsymbol{a}_j\right) + \boldsymbol{p}_j \odot \left(\boldsymbol{E}\boldsymbol{a}_j\right) \odot \boldsymbol{x}_j + \boldsymbol{\varepsilon}_j \end{aligned} \tag{2}$$

where $\mathbf{1}_N \in \mathbb{R}^N$ is a vector of all 1s. The ANC and ASC are imposed, and $\boldsymbol{p}_j$ should meet the constraint, i.e., $\boldsymbol{p}_j < \mathbf{1}_N$. Notably, $\boldsymbol{p}_j$ can be negative, as a pixel might receive extra illumination from 3-D structures outside its instantaneous field of view (IFOV). More details can be found in [15] and [22].

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B. Constrained Optimization Problem Based on the EMLM

The nonconvexity of the objective function and the underdetermined system of equations may cause the EMLM-based unmixing algorithm to converge to undesirable local minima. To address this, incorporating regularizers leveraging HSIs' intrinsic properties into the objective function is a potential approach. Solving the constrained optimization problem in (3) yields accurate unmixing results

$$\begin{aligned}\min_{\boldsymbol{a}_j,\boldsymbol{p}_j}\ & \frac{1}{2}\left\|\left(\mathbf{1}_N-\boldsymbol{p}_j\right)\odot\left(\boldsymbol{E}\boldsymbol{a}_j\right)+\boldsymbol{p}_j\odot\left(\boldsymbol{E}\boldsymbol{a}_j\right)\odot\boldsymbol{x}_j-\boldsymbol{x}_j\right\|_2^2 \\ & +\lambda\left\|\boldsymbol{a}_j\right\|_1+\frac{\tau}{2}\boldsymbol{p}_j^{\top}\boldsymbol{L}\boldsymbol{p}_j \\ \text{s.t. } & \boldsymbol{a}_j\geq 0,\quad \mathbf{1}_R^{\top}\boldsymbol{a}_j=1,\ \boldsymbol{p}_j<\mathbf{1}_N\end{aligned} \tag{3}$$

where λ and τ are the two parameters to balance the contributions of regularizers.

The first term in (3) represents the reconstruction error term based on the EMLM. The second term is the ℓ_1 norm-based sparsity regularizer for abundances. Due to the local spatial distribution of ground covers, endmembers are not likely to simultaneously appear in all pixels. In other words, the abundances of a large part of endmembers in a pixel are actually close to zero, known as the sparsity of abundances. Although the nonlinear intensity varies along with different wavelengths, neighboring bands often exhibit similar multiple scattering, indicating similar nonlinear parameters. The third term corresponds to the Laplacian graph regularizer for nonlinear parameters and is expressed as follows:

$$\frac{1}{2}\boldsymbol{p}_j^{\top}\boldsymbol{L}\boldsymbol{p}_j=\frac{1}{2}\sum_{i=1}^{N}\sum_{k=1}^{N}\left(p_{i,j}-p_{k,j}\right)^2\boldsymbol{W}_{i,k} \tag{4}$$

where $\boldsymbol{L}=\boldsymbol{D}-\boldsymbol{W}$ is a Laplacian matrix, $\boldsymbol{D}$ is a diagonal matrix with the entries $\boldsymbol{D}_{i,i}=\sum_{k=1}^{N}\boldsymbol{W}_{i,k}$, and $\boldsymbol{W}\in\mathbb{R}^{N\times N}$ is a weight matrix defined as

$$\boldsymbol{W}_{i,k}=\begin{cases}e^{-\|\boldsymbol{X}(i,:)-\boldsymbol{X}(k,:)\|_2^2/\theta^2}, & \text{bands } i,k \text{ are neighbors}\\ 0, & \text{others}\end{cases} \tag{5}$$

where $\boldsymbol{X}(i,:)$ is the data vector at the ith band and θ is an empirical parameter set to 1. A larger weight $\boldsymbol{W}_{i,k}$ is assigned to minimize the difference between $p_{i,j}$ and $p_{k,j}$ when $\boldsymbol{X}(i,:)$ and $\boldsymbol{X}(k,:)$ are similar [22].

III. Proposed Method

As shown in Fig. 1, an EMLM-inspired dual-stream network, EMLM-Net, is proposed to provide an interpretable network architecture and fully utilize the spatial information of HSIs. In the basic flowchart of EMLM-Net, two branches are designed to have identical AE-like architectures to facilitate unsupervised training. In this architecture, the encoder part is constructed as a deep unfolding network with K layers, where each network layer corresponds to one iteration of the ADMM for the constrained optimization problem based on the EMLM. The inputs of the encoder are the original HSIs and coarse HSIs generated through superpixel segmentation, accompanied by several pseudo-inputs that function as auxiliary variables and dual variables in the ADMM. The decoder part is derived according to the EMLM and employs the outputs of the encoder, i.e., abundances and nonlinear parameters, to reconstruct the original HSIs and coarse HSIs, respectively.

Furthermore, to fully exploit the spatial information in HSIs, the two branches of EMLM-Net share weights. Since the region-based endmember extraction contributes to obtaining high-quality endmember spectra, the accurate endmembers extracted by the upper branch network positively influence the lower branch network. On the other hand, the estimates of abundances and nonlinear parameters output by the two branches are used to construct the regularizers for the loss function, considering the local spatial similarity inherent in HSIs. After sufficient training, EMLM-Net can yield accurate unmixing results. In the following subsections, the details of EMLM-Net are provided.

A. Unfolding ADMM Into a Neural Network Layer

The ADMM is applied to solve the constrained optimization problem in (3). To simplify notation, the subscript j for $\boldsymbol{x}$, $\boldsymbol{a}$, and $\boldsymbol{p}$ is omitted, as they can be updated pixel by pixel in parallel. By introducing the auxiliary variables $\boldsymbol{v}_1,\boldsymbol{v}_2$, and $\boldsymbol{v}_3$, such that $\boldsymbol{v}_1=\boldsymbol{E}\boldsymbol{a}$, $\boldsymbol{v}_2=\boldsymbol{a}$, $\boldsymbol{v}_3=\boldsymbol{p}$ along with the dual variables $\boldsymbol{d}_1,\boldsymbol{d}_2$, and $\boldsymbol{d}_3$, the augmented Lagrangian for (3) can be derived as

$$\begin{aligned}\arg\min_{\boldsymbol{a},\boldsymbol{p},\boldsymbol{v}_i,\boldsymbol{d}_i}\ & \frac{1}{2}\|\boldsymbol{x}-(\mathbf{1}_N-\boldsymbol{p})\odot\boldsymbol{v}_1-\boldsymbol{p}\odot\boldsymbol{v}_1\odot\boldsymbol{x}\|_2^2 \\ & +\lambda\|\boldsymbol{v}_2\|_1+\frac{\tau}{2}\boldsymbol{v}_3^{\top}\boldsymbol{L}\boldsymbol{v}_3 \\ & +\frac{\mu}{2}\left(\|\boldsymbol{E}\boldsymbol{a}-\boldsymbol{v}_1-\boldsymbol{d}_1\|_2^2+\|\boldsymbol{a}-\boldsymbol{v}_2-\boldsymbol{d}_2\|_2^2\right. \\ & \left.\quad +\|\boldsymbol{p}-\boldsymbol{v}_3-\boldsymbol{d}_3\|_2^2\right) \\ & +\tau_B(\boldsymbol{v}_2)+\tau_C(\boldsymbol{v}_3)\end{aligned} \tag{6}$$

where μ is a penalty parameter with a small positive value, and $\tau_B(\cdot)$ and $\tau_C(\cdot)$ are the two indicator functions to project $\boldsymbol{v}_2$ and $\boldsymbol{v}_3$ onto the set $\{\boldsymbol{v}_2|\boldsymbol{v}_2\geq 0\}$ and $\{\boldsymbol{v}_3|\boldsymbol{v}_3<\mathbf{1}_N\}$. Note that the ASC is not imposed on $\boldsymbol{v}_2$ here. Combining the ℓ_1 norm-based regularizer with the ASC on the same variable may not work, as the ℓ_1 norm will always be one [46]. Unlike BNLSU [22], the EMLM-based unmixing algorithm derived in this article introduces additional auxiliary variables and dual variables for $\boldsymbol{E}\boldsymbol{a}$. This modification enables a straightforward mapping of the algorithm's computational steps to operations in the network layers. The pseudocode for the proposed algorithm is provided in Algorithm 1, and the detailed computation steps can be referred to in Appendix.

When solving the unmixing problem, utilizing the ADMM-based deep unfolding network shows significant advantages over directly applying the ADMM. This approach enables end-to-end learning, adaptively capturing intricate features of different types of data, and reducing the reliance on manually tuned parameters. Therefore, the unmixing performance and generalization get improved. On the other hand, transforming the computational steps of the ADMM into network layers allows parameters learned by the network

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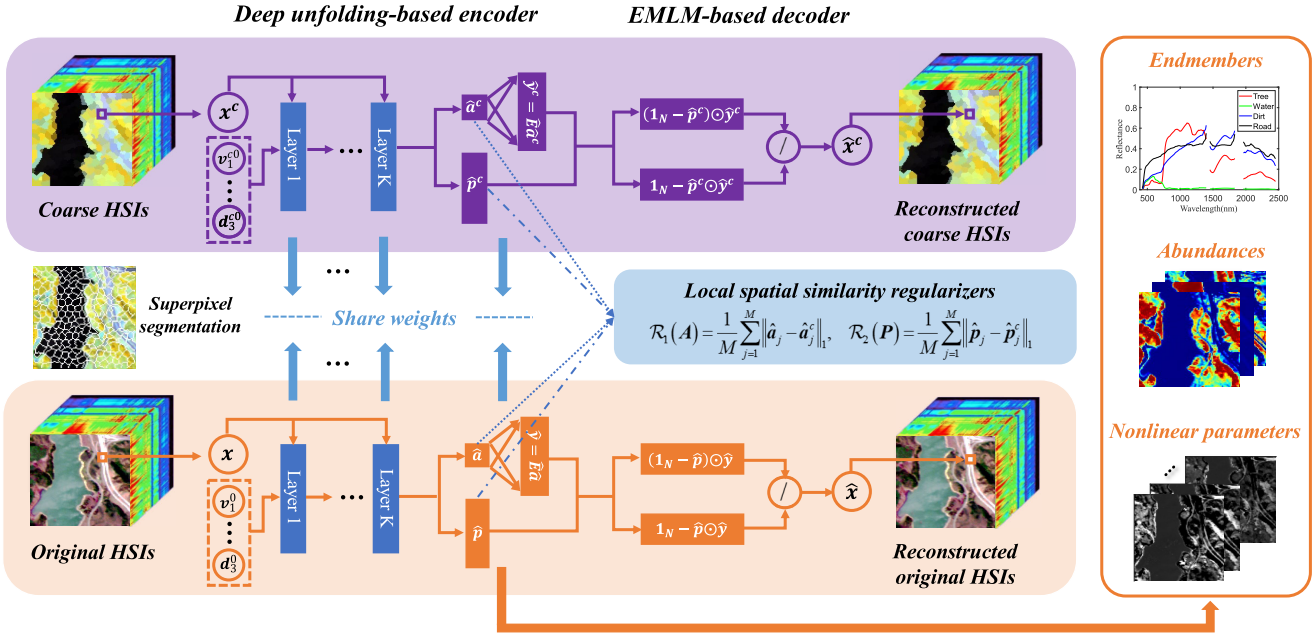


Fig. 1. Flowchart of EMLM-Net.

Algorithm 1 An EMLM-based unmixing method via the ADMM.

Input: Hyperspectral data $\boldsymbol{x} \in \mathbb{R}^N$, endmember matrix $\boldsymbol{E} \in \mathbb{R}^{N\times R}$.

Output: Abundances $\boldsymbol{a} \in \mathbb{R}^R$ and nonlinear parameters $\boldsymbol{p} \in \mathbb{R}^N$.

Set $k=0$, $\mu>0$, $\lambda>0$, and $\tau>0$.
Initialization: $\boldsymbol{a}^0$, $\boldsymbol{p}^0$, and $\boldsymbol{v}_n^0, \boldsymbol{d}_n^0, n=1,2,3$.
While stopping conditions [22] are not met, **do**
  Update $\boldsymbol{a}^{k+1}$ and $\boldsymbol{p}^{k+1}$ with (25) and (27), respectively.
  Update $\boldsymbol{v}_n^{k+1}, n=1,2,3$ with (29), (31), and (33), respectively.
  Update $\boldsymbol{d}_n^{k+1}, n=1,2,3$ with (34)-(36), respectively.
  $k \leftarrow k+1$.
End

to be interpreted as intermediate variables in the problem-solving process, enhancing the network's interpretability. This facilitates the integration of task-relevant prior knowledge, further refining unmixing accuracy. In the following, a detailed explanation will be provided on how the computational steps of the ADMM are mapped to different components of the deep unfolding network layer during the algorithm unfolding process.

1) Components for Updating $\boldsymbol{a}$ and $\boldsymbol{p}$: In the $\boldsymbol{a}$ update component of the $(k+1)$th neural network layer, $\boldsymbol{a}^{k+1}$ is expressed as follows:

$$\begin{aligned}\boldsymbol{a}^{k+1} &= f_a\left(\boldsymbol{v}_1^k, \boldsymbol{v}_2^k, \boldsymbol{d}_1^k, \boldsymbol{d}_2^k; \boldsymbol{\omega}, \boldsymbol{b}\right)\\ &= \boldsymbol{\omega}^\top\left(\boldsymbol{v}_1^k + \boldsymbol{d}_1^k\right) + \boldsymbol{b}^\top\left(\boldsymbol{v}_2^k + \boldsymbol{d}_2^k\right)\end{aligned} \tag{7}$$

where $\boldsymbol{\omega}^\top = (\boldsymbol{E}^\top\boldsymbol{E}+\boldsymbol{I}_{R\times R})^{-1}\boldsymbol{E}^\top$, $\boldsymbol{b}^\top = (\boldsymbol{E}^\top\boldsymbol{E}+\boldsymbol{I}_{R\times R})^{-1}$, and $\boldsymbol{I}_{R\times R}$ is an identity matrix. Furthermore, the learnable parameters $\boldsymbol{\omega}^{k+1} \in \mathbb{R}^{N\times R}$ and $\boldsymbol{b}^{k+1} \in \mathbb{R}^{R\times R}$ are employed in the $(k+1)$th layer, which differ from the original ones $\boldsymbol{\omega}$ and $\boldsymbol{b}$. In this way, greater flexibility is obtained, allowing for better adaptation to the characteristics of the data. The operation performed by the $\boldsymbol{a}$ update component can be considered analogous to a linear layer in a neural network [51].

In the $\boldsymbol{p}$ update component of the $(k+1)$th neural network layer, by utilizing a learnable parameter $\theta^{k+1} \in \mathbb{R}$ that plays a similar role as μ, $\boldsymbol{p}^{k+1}$ is estimated in the following form:

$$\begin{aligned}\boldsymbol{p}^{k+1} &= f_p\left(\boldsymbol{v}_1^k, \boldsymbol{v}_3^k, \boldsymbol{d}_3^k, \boldsymbol{x}; \theta^{k+1}\right)\\ &= \frac{\left(\boldsymbol{x}-\boldsymbol{v}_1^k\right)\odot\left(\boldsymbol{v}_1^k\odot\boldsymbol{x}-\boldsymbol{v}_1^k\right)+\theta^{k+1}\left(\boldsymbol{v}_3^k+\boldsymbol{d}_3^k\right)}{\left(\boldsymbol{v}_1^k\odot\boldsymbol{x}-\boldsymbol{v}_1^k\right)^2+\theta^{k+1}}.\end{aligned} \tag{8}$$

2) Components for Updating $\boldsymbol{v}_1$, $\boldsymbol{v}_2$, and $\boldsymbol{v}_3$: In the $\boldsymbol{v}_1$ update component of the $(k+1)$th neural network layer, $\boldsymbol{v}_1^{k+1}$ is determined by combining $\boldsymbol{a}^{k+1}$, $\boldsymbol{p}^{k+1}$, and $\boldsymbol{d}_1^k$, along with $\boldsymbol{x}$

$$\begin{aligned}\boldsymbol{v}_1^{k+1} &= f_{v_1}\left(\boldsymbol{a}^{k+1}, \boldsymbol{p}^{k+1}, \boldsymbol{d}_1^k, \boldsymbol{x}; \boldsymbol{c}^{k+1}, \eta^{k+1}\right)\\ &= \frac{\eta^{k+1}\left(\mathbf{1}_N-\boldsymbol{p}^{k+1}+\boldsymbol{p}^{k+1}\odot\boldsymbol{x}\right)\odot\boldsymbol{x}+\left(\boldsymbol{c}^{k+1}\right)^\top\boldsymbol{a}^{k+1}-\boldsymbol{d}_1^k}{\eta^{k+1}\left(\mathbf{1}_N-\boldsymbol{p}^{k+1}+\boldsymbol{p}^{k+1}\odot\boldsymbol{x}\right)^2+\mathbf{1}_N}\end{aligned} \tag{9}$$

where $(\boldsymbol{c}^{k+1})^\top \in \mathbb{R}^{N\times R}$ and $\eta^{k+1} \in \mathbb{R}$ are the two learnable parameters that serve as $\boldsymbol{E}$ and $1/\mu$, respectively.

In the $\boldsymbol{v}_2$ update component of the $(k+1)$th neural network layer, by applying a soft-thresholding operation, i.e., $\text{soft}(x,\theta) = \text{sign}(x)\max\{|x|-\theta, 0\}$, followed by a max operation to $\boldsymbol{a}^{k+1}$ and $\boldsymbol{d}_2^k$, $\boldsymbol{v}_2^{k+1}$ is computed as

$$\begin{aligned}\boldsymbol{v}_2^{k+1} &= f_{v_2}\left(\boldsymbol{a}^{k+1}, \boldsymbol{d}_2^k; \rho^{k+1}\right)\\ &= \max\left(\text{soft}\left(\boldsymbol{a}^{k+1}-\boldsymbol{d}_2^k, \rho^{k+1}\right), 0\right)\end{aligned} \tag{10}$$

where $\rho^{k+1} \in \mathbb{R}$ is a learnable parameter that operates akin to λ/μ. The equivalent form of the above equation is provided in the following, where $\text{ReLU}(\cdot)$ is a componentwise rectified linear unit operation. Accordingly, not only can the ANC

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constraint be satisfied, but it also emulates the functionality of the ReLU operation in a typical neural network [46], [51]

$$\boldsymbol{v}_2^{k+1} = \mathrm{ReLU}\left(\boldsymbol{a}^{k+1} - \boldsymbol{d}_2^k - \rho^{k+1}\right). \tag{11}$$

In the $\boldsymbol{v}_3$ update component of the $(k+1)$th neural network layer, the constraint for values less than one is reformulated. $\boldsymbol{v}_3^{k+1}$ is calculated using $\boldsymbol{p}^{k+1}$, $\boldsymbol{d}_3^k$, and a learnable parameter $(\boldsymbol{g}^{k+1})^\top \in \mathbb{R}^{N\times N}$ that operates as $(\tau \boldsymbol{L} + \mu \boldsymbol{I}_{N\times N})^{-1}\mu$, where $\boldsymbol{I}_{N\times N}$ is an identity matrix

$$\begin{aligned}\boldsymbol{v}_3^{k+1} &= f_{v_3}\left(\boldsymbol{p}^{k+1}, \boldsymbol{d}_3^k; \boldsymbol{g}^{k+1}\right) \\ &= \mathbf{1}_N - \max\left(\mathbf{1}_N - \left(\boldsymbol{g}^{k+1}\right)^\top \left(\boldsymbol{p}^{k+1} - \boldsymbol{d}_3^k\right), 0\right).\end{aligned} \tag{12}$$

The operation executed by the $\boldsymbol{v}_3$ update component can be alternatively expressed as

$$\boldsymbol{v}_3^{k+1} = \mathbf{1}_N - \mathrm{ReLu}\left(\mathbf{1}_N - \left(\boldsymbol{g}^{k+1}\right)^\top \left(\boldsymbol{p}^{k+1} - \boldsymbol{d}_3^k\right)\right). \tag{13}$$

3) Components for Updating $\boldsymbol{d}_1$, $\boldsymbol{d}_2$, and $\boldsymbol{d}_3$: In the $\boldsymbol{d}_1$ update component of the $(k+1)$th neural network layer, $\boldsymbol{d}_1^{k+1}$ is derived as

$$\begin{aligned}\boldsymbol{d}_1^{k+1} &= f_{d_1}\left(\boldsymbol{a}^{k+1}, \boldsymbol{v}_1^{k+1}, \boldsymbol{d}_1^k; \boldsymbol{h}^{k+1}, \sigma_1^{k+1}\right) \\ &= \sigma_1^{k+1}\boldsymbol{d}_1^k - \left(\left(\boldsymbol{h}^{k+1}\right)^\top \boldsymbol{a}^{k+1} - \boldsymbol{v}_1^{k+1}\right)\end{aligned} \tag{14}$$

where $(\boldsymbol{h}^{k+1})^\top \in \mathbb{R}^{N\times R}$ is a learnable parameter that functions as $\boldsymbol{E}$, and $\sigma_1^{k+1} \in \mathbb{R}$ is another learnable parameter that can be regarded as a dynamic step size, providing additional flexibility to adapt to the characteristics of the unmixing problem. Note that we multiply the learnable parameter σ_1^{k+1} with the parameter $\boldsymbol{d}_1^k$, rather than with the second term in (14), to prevent coupling between the learnable parameters $\boldsymbol{h}^{k+1}$ and σ_1^{k+1}, which could pose quite a challenge to the optimization.

In the $\boldsymbol{d}_2$ update component and $\boldsymbol{d}_3$ update component of the $(k+1)$th neural network layer, by introducing learnable parameters σ_2^{k+1} and σ_3^{k+1}, which act similar to step sizes, $\boldsymbol{d}_2^{k+1}$ and $\boldsymbol{d}_3^{k+1}$ can be reexpressed as

$$\begin{aligned}\boldsymbol{d}_2^{k+1} &= f_{d_2}\left(\boldsymbol{a}^{k+1}, \boldsymbol{v}_2^{k+1}, \boldsymbol{d}_2^k; \sigma_2^{k+1}\right) \\ &= \boldsymbol{d}_2^k - \sigma_2^{k+1}\left(\boldsymbol{a}^{k+1} - \boldsymbol{v}_2^{k+1}\right)\end{aligned} \tag{15}$$

$$\begin{aligned}\boldsymbol{d}_3^{k+1} &= f_{d_3}\left(\boldsymbol{p}^{k+1}, \boldsymbol{v}_3^{k+1}, \boldsymbol{d}_3^k; \sigma_3^{k+1}\right) \\ &= \boldsymbol{d}_3^k - \sigma_3^{k+1}\left(\boldsymbol{p}^{k+1} - \boldsymbol{v}_3^{k+1}\right).\end{aligned} \tag{16}$$

As shown in Table I, the computational steps of the ADMM are mapped to different components of the deep unfolding network layer, when performing algorithm unfolding.

4) The Structure of a Neural Network Layer: Eight components for updating different variables, with specific network structures illustrated in Fig. 2(a)–(h), are combined to form a neural network layer in Fig. 3. Each layer corresponds to one iteration of the ADMM, but with the layer-specific learnable parameters $\boldsymbol{\Theta}^k = \{\boldsymbol{\omega}^k, \boldsymbol{b}^k, \boldsymbol{c}^k, \boldsymbol{g}^k, \boldsymbol{h}^k, \theta^k, \eta^k, \rho^k, \sigma_1^k, \sigma_2^k, \sigma_3^k\}$, $\forall k \in [1, K]$ instead of hand-coded ones. Compared to the numerous iterations required by the ADMM, this parameterization can achieve desired accuracy with fewer layers [43], [46].

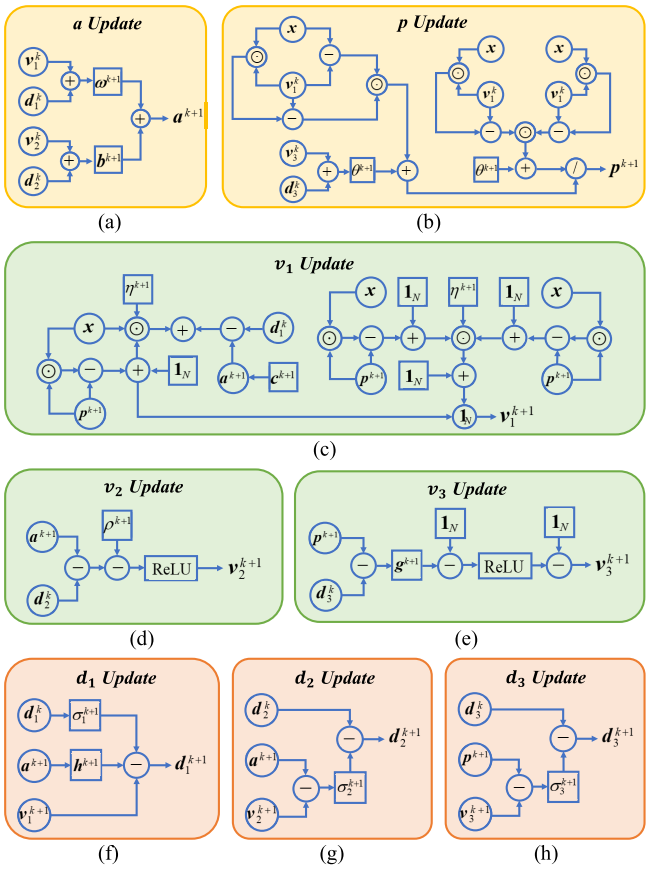


Fig. 2. Network structures corresponding to the eight components (a)–(h) for updating different variables within a neural network layer.

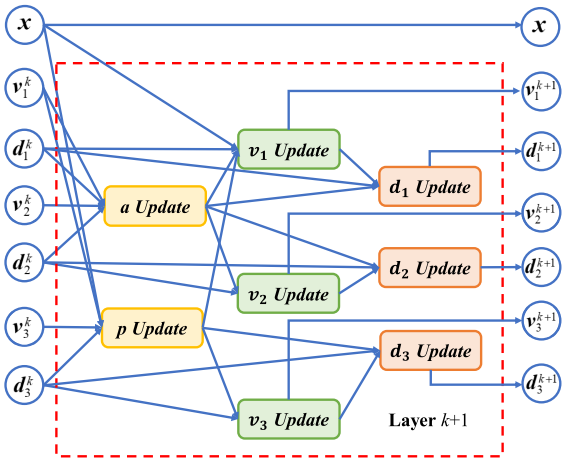


Fig. 3. Structure of a neural network layer that simulates one iteration of the ADMM.

B. AE-Like Network Architecture

By concatenating K iterative blocks presented in Fig. 3, a feedforward neural network is formulated to mimic K iterations of the ADMM. The inputs are passed through a K-layer deep unfolding network to estimate abundances and nonlinear parameters. This process resembles an encoder network. A decoder based on the EMLM is then incorporated, creating an AE-like network. Notably, the weights of the decoder part serve as the estimation of endmember spectra.

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TABLE I
COMPUTATIONAL STEPS OF THE ADMM ARE MAPPED TO DIFFERENT COMPONENTS OF THE DEEP UNFOLDING NETWORK LAYER

Updated variables	*Calculation formulas in the ADMM*	*Components of the deep unfolding network layer*
$\boldsymbol{a}$	$\boldsymbol{a}^{k+1} = \left(\boldsymbol{E}^\top \boldsymbol{E} + \boldsymbol{I}_{R\times R}\right)^{-1}\left(\boldsymbol{E}^\top\left(\boldsymbol{v}_1^k + \boldsymbol{d}_1^k\right) + \left(\boldsymbol{v}_2^k + \boldsymbol{d}_2^k\right)\right)$	$\begin{aligned}\boldsymbol{a}^{k+1} &= f_a\left(\boldsymbol{v}_1^k, \boldsymbol{v}_2^k, \boldsymbol{d}_1^k, \boldsymbol{d}_2^k; \boldsymbol{\omega}, \boldsymbol{b}\right)\\ &= \boldsymbol{\omega}^\top\left(\boldsymbol{v}_1^k + \boldsymbol{d}_1^k\right) + \boldsymbol{b}^\top\left(\boldsymbol{v}_2^k + \boldsymbol{d}_2^k\right)\end{aligned}$
$\boldsymbol{p}$	$\boldsymbol{p}^{k+1} = \dfrac{\left(\boldsymbol{x} - \boldsymbol{v}_1^k\right)\odot\left(\boldsymbol{v}_1^k \odot \boldsymbol{x} - \boldsymbol{v}_1^k\right) + \mu\left(\boldsymbol{v}_3^k + \boldsymbol{d}_3^k\right)}{\left(\boldsymbol{v}_1^k \odot \boldsymbol{x} - \boldsymbol{v}_1^k\right)^2 + \mu}$	$\begin{aligned}\boldsymbol{p}^{k+1} &= f_p\left(\boldsymbol{v}_1^k, \boldsymbol{v}_3^k, \boldsymbol{d}_3^k, \boldsymbol{x}; \theta^{k+1}\right)\\ &= \dfrac{\left(\boldsymbol{x} - \boldsymbol{v}_1^k\right)\odot\left(\boldsymbol{v}_1^k \odot \boldsymbol{x} - \boldsymbol{v}_1^k\right) + \theta^{k+1}\left(\boldsymbol{v}_3^k + \boldsymbol{d}_3^k\right)}{\left(\boldsymbol{v}_1^k \odot \boldsymbol{x} - \boldsymbol{v}_1^k\right)^2 + \theta^{k+1}}\end{aligned}$
$\boldsymbol{v}_1$	$\boldsymbol{v}_1^{k+1} = \dfrac{1/\mu\left(\boldsymbol{1}_N - \boldsymbol{p}^{k+1} + \boldsymbol{p}^{k+1}\odot\boldsymbol{x}\right)\odot\boldsymbol{x} + \boldsymbol{E}\boldsymbol{a}^{k+1} - \boldsymbol{d}_1^k}{1/\mu\left(\boldsymbol{1}_N - \boldsymbol{p}^{k+1} + \boldsymbol{p}^{k+1}\odot\boldsymbol{x}\right)^2 + \boldsymbol{1}_N}$	$\begin{aligned}\boldsymbol{v}_1^{k+1} &= f_{v_1}\left(\boldsymbol{a}^{k+1}, \boldsymbol{p}^{k+1}, \boldsymbol{d}_1^k, \boldsymbol{x}; \boldsymbol{c}^{k+1}, \eta^{k+1}\right)\\ &= \dfrac{\eta^{k+1}\left(\boldsymbol{1}_N - \boldsymbol{p}^{k+1} + \boldsymbol{p}^{k+1}\odot\boldsymbol{x}\right)\odot\boldsymbol{x} + \left(\boldsymbol{c}^{k+1}\right)^\top\boldsymbol{a}^{k+1} - \boldsymbol{d}_1^k}{\eta^{k+1}\left(\boldsymbol{1}_N - \boldsymbol{p}^{k+1} + \boldsymbol{p}^{k+1}\odot\boldsymbol{x}\right)^2 + \boldsymbol{1}_N}\end{aligned}$
$\boldsymbol{v}_2$	$\boldsymbol{v}_2^{k+1} = \max\left(\mathrm{soft}\left(\boldsymbol{a}^{k+1} - \boldsymbol{d}_2^k, \lambda/\mu\right), 0\right)$	$\begin{aligned}\boldsymbol{v}_2^{k+1} &= f_{v_2}\left(\boldsymbol{a}^{k+1}, \boldsymbol{d}_2^k; \rho^{k+1}\right)\\ &= \max\left(soft\left(\boldsymbol{a}^{k+1} - \boldsymbol{d}_2^k, \rho^{k+1}\right), 0\right)\end{aligned}$
$\boldsymbol{v}_3$	$\boldsymbol{v}_3^{k+1} = \min\left(\left(\tau\boldsymbol{L} + \mu\boldsymbol{I}_{N\times N}\right)^{-1}\left(\mu\left(\boldsymbol{p}^{k+1} - \boldsymbol{d}_3^k\right)\right), \boldsymbol{1}_N\right)$	$\begin{aligned}\boldsymbol{v}_3^{k+1} &= f_{v_3}\left(\boldsymbol{p}^{k+1}, \boldsymbol{d}_3^k; \boldsymbol{g}^{k+1}\right)\\ &= \boldsymbol{1}_N - \max\left(\boldsymbol{1}_N - \left(\boldsymbol{g}^{k+1}\right)^\top\left(\boldsymbol{p}^{k+1} - \boldsymbol{d}_3^k\right), 0\right)\end{aligned}$
$\boldsymbol{d}_1$	$\boldsymbol{d}_1^{k+1} = \boldsymbol{d}_1^k - \left(\boldsymbol{E}\boldsymbol{a}^{k+1} - \boldsymbol{v}_1^{k+1}\right)$	$\begin{aligned}\boldsymbol{d}_1^{k+1} &= f_{d_1}\left(\boldsymbol{a}^{k+1}, \boldsymbol{v}_1^{k+1}, \boldsymbol{d}_1^k; \boldsymbol{h}^{k+1}, \sigma_1^{k+1}\right)\\ &= \sigma_1^{k+1}\boldsymbol{d}_1^k - \left(\left(\boldsymbol{h}^{k+1}\right)^\top\boldsymbol{a}^{k+1} - \boldsymbol{v}_1^{k+1}\right)\end{aligned}$
$\boldsymbol{d}_2$	$\boldsymbol{d}_2^{k+1} = \boldsymbol{d}_2^k - \left(\boldsymbol{a}^{k+1} - \boldsymbol{v}_2^{k+1}\right)$	$\begin{aligned}\boldsymbol{d}_2^{k+1} &= f_{d_2}\left(\boldsymbol{a}^{k+1}, \boldsymbol{v}_2^{k+1}, \boldsymbol{d}_2^k; \sigma_2^{k+1}\right)\\ &= \boldsymbol{d}_2^k - \sigma_2^{k+1}\left(\boldsymbol{a}^{k+1} - \boldsymbol{v}_2^{k+1}\right)\end{aligned}$
$\boldsymbol{d}_3$	$\boldsymbol{d}_3^{k+1} = \boldsymbol{d}_3^k - \left(\boldsymbol{p}^{k+1} - \boldsymbol{v}_3^{k+1}\right)$	$\begin{aligned}\boldsymbol{d}_3^{k+1} &= f_{d_3}\left(\boldsymbol{p}^{k+1}, \boldsymbol{v}_3^{k+1}, \boldsymbol{d}_3^k; \sigma_3^{k+1}\right)\\ &= \boldsymbol{d}_3^k - \sigma_3^{k+1}\left(\boldsymbol{p}^{k+1} - \boldsymbol{v}_3^{k+1}\right)\end{aligned}$

1) Encoder: In the encoder network as depicted in Fig. 4, along with the input pixel $\boldsymbol{x}$, pseudo-inputs like $\boldsymbol{v}_i^0$ and $\boldsymbol{d}_i^0$ $(i = 1, 2, 3)$ are introduced to simulate the iterative process of the ADMM. The outputs of the encoder network are derived from the $\boldsymbol{v}_2$ update component and the $\boldsymbol{v}_3$ update component in the Kth iterative block. Given that the optimization problem only considers the ANC constraint, the final output of the $\boldsymbol{v}_2$ update component is normalized using l_1 normalization to satisfy the ASC constraint. Denoting the prenormalized network output as $\boldsymbol{v}_2^K$, the normalized output $\hat{\boldsymbol{a}}$ is computed as $\hat{\boldsymbol{a}}_i = (\boldsymbol{v}_2^K)_i / \sum_j (\boldsymbol{v}_2^K)_j$. On the other hand, the output of the $\boldsymbol{v}_3$ update component is regarded as the nonlinear parameter estimate, denoted as $\hat{\boldsymbol{p}}$.

2) Decoder: Using the outputs of the encoder, an EMLM-based decoder network can be readily constructed. The mathematical expression of the EMLM in (2) allows us to derive the following equation:

$$\hat{\boldsymbol{x}} = \frac{\left(\boldsymbol{1}_N - \hat{\boldsymbol{p}}\right)\odot\hat{\boldsymbol{y}}}{\boldsymbol{1}_N - \hat{\boldsymbol{p}}\odot\hat{\boldsymbol{y}}} \tag{17}$$

where $\hat{\boldsymbol{x}}$ is the reconstructed spectrum, and $\hat{\boldsymbol{y}} = \hat{\boldsymbol{E}}\hat{\boldsymbol{a}}$ is the reconstructed linear component. The matrix $\hat{\boldsymbol{E}}$ corresponds to the weights of the fully connected layer connecting $\hat{\boldsymbol{a}}$ and $\hat{\boldsymbol{y}}$, serving as the endmember spectra estimate.

By combining the deep unfolding-based encoder with the EMLM-based decoder, an AE-like unmixing network can be built in Fig. 4. Overall, this network has several advantages: first, compared to conventional AE-based unmixing networks, it exhibits stronger interpretability, as its encoder part is derived from unfolding the ADMM. Second, an EMLM-based decoder is introduced, well-suited for addressing real-world challenges arising from the prevalent nonlinear mixing effects.

C. Dual-Stream Unmixing Network

To incorporate spatial information of HSIs into the network, the AE-like unmixing network presented in Fig. 4 is expanded into a dual-stream unmixing network. This network simultaneously processes the original HSIs and the superpixel-averaging-based coarse HSIs. It adopts a weight-sharing strategy and introduces regularizers in the loss function to capture the local spatial similarity of HSIs. The implementation of this network architecture significantly enhances the robustness and generalization capabilities of the unmixing algorithm, yielding more reliable and accurate results.

1) Overall Network Architecture: The comprehensive network architecture is provided in Fig. 1. The upper and lower branch networks mirror the structure in Fig. 4, differing only in inputs: one processes a pixel $\boldsymbol{x}$ from the original HSIs, while the other takes a pixel $\boldsymbol{x}^c$ from the coarse HSIs. The coarse HSIs are obtained after replacing the original pixels within superpixels with their average spectrum. These superpixels, representing local spatial neighborhoods, are generated through the segmentation of the original HSIs via the SLIC [47]. By utilizing both $\boldsymbol{x}$ and $\boldsymbol{x}^c$ as network inputs, along with their respective pseudo-inputs, and traversing a K-layer network, the abundances and nonlinear parameters can be estimated accordingly. They are then fed into the EMLM-based decoders, leading to the reconstruction of the pixel spectra, $\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{x}}^c$.

2) Weight-Sharing Strategy: To establish an effective association between the upper and lower branch networks,

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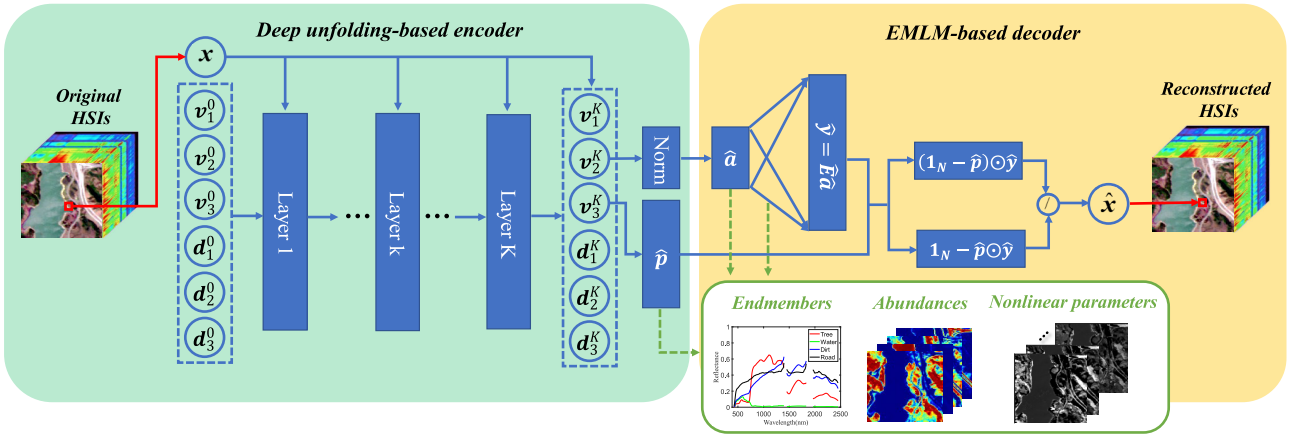


Fig. 4. AE-like unmixing network using the deep unfolding.

a weight-sharing strategy is adopted. Given that both branch networks perform unmixing tasks in the same scene, if the network is viewed as a function, it is natural to employ the same function for unmixing the original HSIs and the corresponding coarse HSIs [38]. Therefore, the encoders of both branch networks share weights. On the other hand, the decoder weights in both branch networks possess the same physical meaning, representing estimated endmembers. The upper branch network is dedicated to region-based endmember extraction [49], [50]. Compared to the global-based method, this method utilizes the average spectrum of each region as a candidate pixel, effectively capturing spectral characteristics of individual regions and demonstrating robustness to noise. Besides, by reducing the reliance on training the network for all pixels, it enhances the consistency in learning weights for the same class of endmembers, mitigating the impact of spectral variability. The upper branch network yields accurate endmember spectra and positively influences the lower branch network through the weight sharing, consequently improving the accuracy of estimating abundances and nonlinear parameters in the lower branch network.

3) Regularizers: As stated in [17] and [24], pixels within a local region should not only share similar abundances but also exhibit similar nonlinear parameters. To leverage and integrate the inherent spatial information of HSIs into the network, we propose to introduce the local spatial similarity regularizers for abundances and nonlinear parameters, denoted as $\mathcal{R}_1(\boldsymbol{A})$ and $\mathcal{R}_2(\boldsymbol{P})$, into the loss function. Their mathematical expressions are as follows:

$$\mathcal{R}_1(\boldsymbol{A}) = \frac{1}{M}\sum_{j=1}^{M}\left\|\hat{\boldsymbol{a}}_j - \hat{\boldsymbol{a}}_j^c\right\|_1 \tag{18}$$

$$\mathcal{R}_2(\boldsymbol{P}) = \frac{1}{M}\sum_{j=1}^{M}\left\|\hat{\boldsymbol{p}}_j - \hat{\boldsymbol{p}}_j^c\right\|_1 \tag{19}$$

where $\hat{\boldsymbol{a}}_j$ and $\hat{\boldsymbol{p}}_j$ denote the abundances and nonlinear parameters estimated from the jth pixel $\boldsymbol{x}_j$ in the original HSIs through the lower branch network. $\hat{\boldsymbol{a}}_j^c$ and $\hat{\boldsymbol{p}}_j^c$ denote the abundances and nonlinear parameters estimated for the jth pixel $\boldsymbol{x}_j^c$ in coarse HSIs through the upper branch network. By employing the l_1 norm-based regularizers to enforce the similarity in both the abundance estimates and nonlinear parameter estimates of the two branch networks, the priors of local spatial similarity in HSIs can be effectively incorporated into the unmixing network.

4) Loss Function: The scale-invariant spectral angle distance (SAD) defined in (20) is employed as the reconstruction metric for pixels in both upper and lower branch networks

$$\mathrm{SAD}(\boldsymbol{x}, \hat{\boldsymbol{x}}) = \arccos\left(\frac{\boldsymbol{x}^\top \hat{\boldsymbol{x}}}{\|\boldsymbol{x}\|_2 \|\hat{\boldsymbol{x}}\|_2}\right) \tag{20}$$

where $\boldsymbol{x}$ and $\hat{\boldsymbol{x}}$ are the original pixel and its reconstruction, respectively. Then, $\mathcal{R}_1(\boldsymbol{A})$ and $\mathcal{R}_2(\boldsymbol{P})$ are incorporated to formulate the loss function, which can be expressed as follows:

$$\mathrm{Loss} = \frac{1}{M}\sum_{j=1}^{M}\mathrm{SAD}(\boldsymbol{x}_j, \hat{\boldsymbol{x}}_j) + \frac{1}{M}\sum_{j=1}^{M}\mathrm{SAD}(\boldsymbol{x}_j^c, \hat{\boldsymbol{x}}_j^c) + \beta\mathcal{R}_1(\boldsymbol{A}) + \gamma\mathcal{R}_2(\boldsymbol{P}) \tag{21}$$

where β and γ are the two tradeoff parameters that balance the data reconstruction terms and regularizers.

D. Implementation Details

To initialize the learnable parameters in different update components, the original parameters involved in the ADMM algorithm are utilized. In this article, the endmember matrix $\boldsymbol{E}$ is estimated from coarse HSIs via the VCA [5]. The Laplacian matrix $\boldsymbol{L}$ is computed as in (5), and the parameters λ, τ, and μ are all set as the default values of 10^{-3}. In the $\boldsymbol{a}$ update component of each layer, the parameters $(\boldsymbol{\omega}^k)^\top$ and $(\boldsymbol{b}^k)^\top$ are initialized by $(\boldsymbol{E}^\top\boldsymbol{E} + \boldsymbol{I}_{R\times R})^{-1}\boldsymbol{E}^\top$ and $(\boldsymbol{E}^\top\boldsymbol{E} + \boldsymbol{I}_{R\times R})^{-1}$, respectively. The parameter θ^k in the $\boldsymbol{p}$ update component of each layer is initialized by μ. The parameters $(\boldsymbol{c}^k)^\top$ and η^k in the $\boldsymbol{v}_1$ update component of each layer are initialized by $\boldsymbol{E}$ and $1/\mu$, respectively. The parameter ρ^k in the $\boldsymbol{v}_2$ update component of each layer is initialized by λ/μ. The parameter $(\boldsymbol{g}^k)^\top$ in the $\boldsymbol{v}_3$ update component of each layer is initialized by $(\tau\boldsymbol{L} + \mu\boldsymbol{I}_{N\times N})^{-1}\mu$. The parameters $(\boldsymbol{h}^k)^\top$ and σ_1^k in the $\boldsymbol{d}_1$ update component of each layer are initialized by $\boldsymbol{E}$ and 1, respectively. The parameters σ_2^k and σ_3^{k+1} in the $\boldsymbol{d}_2$ update component and $\boldsymbol{d}_3$ update component of each layer

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are initialized by 1. Besides, the pseudo-inputs of both upper and lower branches are set as zero matrices of different sizes.

According to [46], the number of learnable parameters in the network architecture can serve as a proxy for measuring the network complexity. In the kth layer of EMLM-Net, the number of learnable parameters is $(N^2 + 3NR + R^2 + 6)$, where N^2 corresponds to the matrix $\boldsymbol{g}^k$, $3NR$ to the three matrices $\boldsymbol{\omega}^k$, $\boldsymbol{c}^k$, and $\boldsymbol{h}^k$, R^2 to the matrix $\boldsymbol{b}^k$, and the remaining ones to the six scalars θ^k, η^k, ρ^k, σ_1^k, σ_2^k, and σ_3^k. Since the learnable parameters of EMLM-Net are shared within network branches but not between different network layers, the total number of learnable parameters is $(N^2 + 3NR + R^2 + 6)K + NR$, where K is the layer number of networks and NR corresponds to the weight matrix treated as the estimated endmembers.

IV. Experiments

In this work, the root mean square error (RMSE) and the mean SAD (MSAD) are used to evaluate the accuracy of abundance estimation and endmember extraction, respectively

$$\text{RMSE} = \sqrt{\frac{1}{RM}\sum_{j=1}^{M}\|\boldsymbol{a}_j - \hat{\boldsymbol{a}}_j\|_2^2} \tag{22}$$

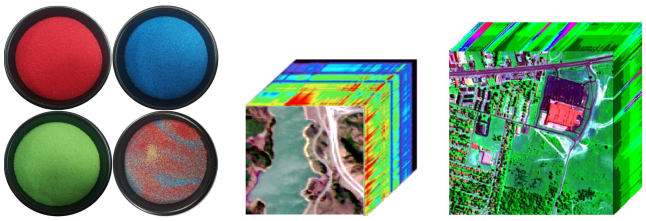
$$\text{MSAD} = \frac{1}{R}\sum_{i=1}^{R}\arccos\left(\frac{\boldsymbol{e}^T\hat{\boldsymbol{e}}_i}{\|\boldsymbol{e}_i\|_2\|\hat{\boldsymbol{e}}_i\|_2}\right) \tag{23}$$

where $\hat{\boldsymbol{a}}_j$ and $\hat{\boldsymbol{e}}_i$ denote the estimated abundance vector for the jth pixel and the ith endmember spectrum, respectively. RMSE and MSAD can quantify the degree of approximation between the estimated and real values. Their smaller values indicate higher accuracy.

A. Hyperspectral Datasets

Considering common nonlinear mixing scenarios, including intimate mixtures at the microscopic scale and multilayer mixtures with large 3-D geometric structures, this study conducted experiments using one laboratory-created dataset and two real datasets. These datasets contain significant nonlinearity and variabilities. Conducting experiments on them sufficiently demonstrates the applicability of the proposed method. In a series of literatures [42], [46], [50], these datasets have been employed to assess the performance of different unmixing algorithms.

1) Laboratory-Created Dataset: Acquired through a push-broom imaging spectrometer, this dataset comprises 256 spectral bands covering visible and near-infrared wavelengths from 400 to 1000 nm, featuring a spectral resolution up to 0.58 nm. As shown in Fig. 5(a), different colors of quartz sands with a uniform size are treated as pure materials. By mixing the quartz sands, a composite-colored quartz sand sample is generated with a spatial pattern. Leveraging the spatial resolution ratio, a high-resolution RGB image aligned with hyperspectral pixels is associated with the scene. Ground truth is then obtained by assessing the proportions of each colored sand in low-resolution hyperspectral pixels, aided by the RGB image. In the experiments, we focused on a 60 × 60 subimage extracted from the center of the image. Further information regarding this laboratory-created data can be found in [52].



(a) (b) (c)

Fig. 5. Hyperspectral datasets used in the experiments. (a) Laboratory-created dataset. (b) Jasper Ridge dataset. (c) Urban dataset.

2) Real Dataset 1 (Jasper Ridge): Acquired by analytical imaging and geophysics (AIG), this dataset contains 100 × 100 pixels, as shown in Fig. 5(b). Each pixel spectrum is captured across 224 channels ranging from 380 to 2500 nm, with a spectral resolution up to 9.46 nm. After excluding channels affected by water vapor and atmospheric conditions, 198 channels are finally retained for analysis. In this scene, four endmembers are considered, including tree, water, soil, and road. Please refer to [53] for more details regarding this dataset and the associated ground truth references.

3) Real Dataset 2 (Urban): This image is captured by the hyperspectral digital imagery collection experiment (HYDICE), and it has 307 × 307 pixels, as shown in Fig. 5(c). It consists of 210 spectral bands ranging from 400 to 2500 nm, with a spectral resolution up to 10 nm. After removing the contaminated channels, 162 channels are remained. In this scene, four main endmembers are observed: asphalt, grass, tree, and roof. More details about this dataset and the reference ground truths can also be found in [53].

B. Parameter Analysis

While conducting experiments on simulated datasets helps validate the performance of unmixing methods, it is important to recognize that the generation of simulated datasets relies on a specific spectral mixing model. As a result, algorithms based on the model used for generating data are inclined to achieve better unmixing accuracy, leading to unfair assessments. Besides, the spectral data usually tend to suffer from various degradations, noise effects, or variabilities in the process of imaging, making them challenging to simulate artificially. Even if the unmixing method demonstrates excellent performance on some simulated datasets, it does not necessarily guarantee its effectiveness in addressing various variabilities in real-world scenarios. Given these, we refrain from designing experiments on simulated datasets, a common practice in literatures related to unmixing. Instead, in the following experiments, the Jasper Ridge dataset is employed as a case study to analyze the unmixing performance of the proposed network.

Furthermore, to ensure the reproducibility of experimental results, the default settings for the parameters involved in the proposed unmixing network are provided. The unmixing network is trained using the Adam optimizer. The initial learning rate is set to 10^{-5}, with the number of decay steps

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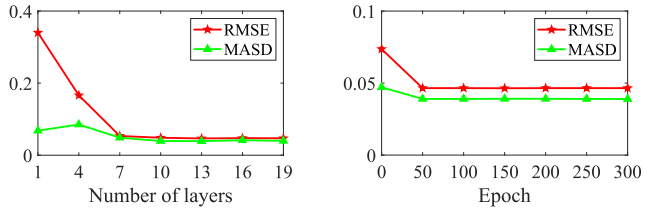


Fig. 6. Unmixing accuracy analysis under various conditions. (a) Different numbers of network layers. (b) Different numbers of epochs.

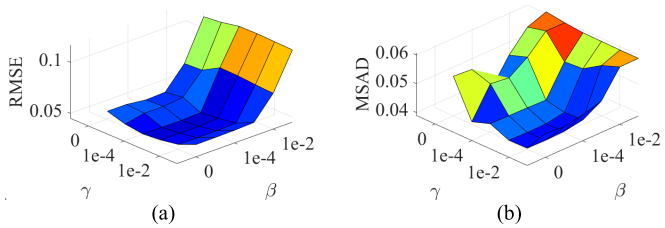


Fig. 7. Variations in unmixing accuracy under different regularizer parameter settings. (a) RMSE. (b) MSAD.

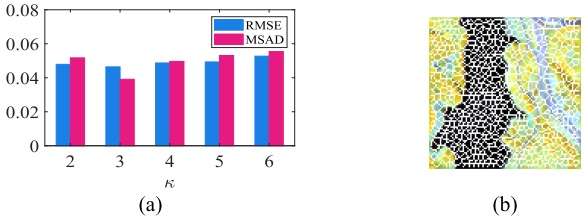


Fig. 8. Impact of superpixel parameters on unmixing accuracy. (a) Variations in RMSE and MSAD for different κ. (b) Generated superpixels.

being 50 and the decay rate being 0.95. The batch size is set to 64, and the number of training epochs is set to 50. Subsequently, we will focus on analyzing the impact of critical hyperparameters on the unmixing accuracy of the proposed method.

1) Impact of Different Numbers of Layers: To evaluate the unmixing accuracy of the proposed method with varying numbers of layers, we change the number of layers while keeping the default experimental settings as mentioned above. In Fig. 6(a), as the number of layers increases, the unmixing accuracy gradually improves. However, further increasing the number of layers does not significantly impact unmixing accuracy, which differs from the behavior of traditional neural networks that typically benefit from a deeper structure. According to [46], this observation can be attributed to the strong inductive bias inherent in the unfolding network architecture, allowing good unmixing accuracy to be achieved with a small number of layers. For the subsequent experiments, the number of network layers is fixed at 13.

2) Impact of Different Numbers of Epochs: Next, the relationship between the unmixing accuracy of the proposed method and the number of training epochs is studied. We set the number of training epochs between 0 and 300, while keeping the remaining parameters unchanged. In Fig. 6(b), the curves depicting the variation of RMSE and MSAD obtained by the proposed method with respect to the number of training epochs are illustrated. Clearly, the proposed method demonstrates rapid convergence, achieving stable accuracy after approximately 50 epochs.

3) Impact of Different Regularizer Parameters: The loss function in (21) involves two hyperparameters β and γ. To determine their values, the variations of RMSE and MSAD obtained by the proposed method under different parameter values are shown in Fig. 7(a) and (b), respectively. Overall, RMSE and MSAD exhibit relatively low sensitivity to the changes of parameters β and γ. Consequently, we set β to 10^{-5} and γ to 10^{-3} and keep the abovementioned parameter settings consistent across all experiments.

4) Impact of Different Superpixel Parameters: At last, the impact of superpixel parameters on unmixing accuracy is analyzed. There are two parameters κ_s and κ involved in the SLIC, where κ_s weighs the relative importance of spectral and spatial similarities, and κ controls the average size of superpixels. As suggested in [48], spectral proximity contributes more to similarity measurement. Thus, κ_s is fixed at 0.3. With κ ranging from 2 to 6, its effect on RMSE and MSAD is depicted in Fig. 8(a). Clearly, the unmixing accuracy of the proposed method remains relatively insensitive to variations in κ within a substantial range. Therefore, κ is set to 3 for subsequent experiments. In Fig. 8(b), the generated superpixels are illustrated when $\kappa = 3$.

TABLE II
RMSE AND MSAD VALUES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS FOR ABLATION EXPERIMENTS

Methods	Single-stream	Dual-stream	$\mathcal{R}_1(\boldsymbol{A})$	$\mathcal{R}_2(\boldsymbol{P})$	RMSE	MSAD
EMLM -Net (S)	√	×	×	×	0.0715 ±0.0117	0.0939 ±0.0111
EMLM -Net (D)	×	√	×	×	0.0469 ±0.0073	0.0463 ±0.0110
EMLM -Net (DR)	×	√	√	×	0.0456 ±0.0061	0.0449 ±0.0092
EMLM -Net	×	√	√	√	**0.0409 ±0.0052**	**0.0369 ±0.0044**

C. Ablation Experiments

To clarify the specific contributions of the innovations introduced in this article, i.e., the dual-stream network structure and the regularizers expressing HSIs' local spatial similarity, to the improvement of unmixing accuracy, it is imperative to conduct ablation experiments. For ease of expression, the single-stream network depicted in Fig. 4 is referred to as EMLM-Net (S), the dual-stream network without any regularizers as EMLM-Net (D), and the dual-stream network with the regularizer on the abundances as EMLM-Net (DR), where the abundance regularizer parameter was fine-tuned and set to $1e^{-4}$. Using the same default parameter settings, these methods and EMLM-Net were run five times. In Table II, the mean and standard deviation of RMSE and MSAD obtained by them are reported, with the best results highlighted in bold.

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TABLE III
RMSE AND MSAD VALUES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS ON THE LABORATORY-CREATED DATASET

Metrics	V/F	uDAS	ADMMNet	EGU-Net	DMBU	HapkeCNN	3D-CNNAE	BNLSU	EMLM-Net
RMSE	0.2616 ±0.0132	0.2585 ±0.0125	0.2306 ±0.0139	0.2534 ±0.0081	0.2609 ±0.0125	0.2650 ±0.0167	0.2221 ±0.0052	0.2497 ±0.0119	**0.2008 ±0.0014**
MSAD	0.1867 ±0.0025	0.1826 ±0.0076	0.1815 ±0.0032	0.1871 ±0.0028	0.1868 ±0.0026	0.1891 ±0.0273	0.1977 ±0.0051	—	**0.1393 ±0.0028**

From the experimental results shown in Table II, we can have the following conclusions. First, compared to EMLM-Net (S), the unmixing method with a dual-stream network structure exhibits a significant improvement in unmixing accuracy, with an increase of at least 34% in RMSE and 54% in MSAD. This clearly demonstrates that the use of a dual-stream network structure substantially enhances unmixing accuracy. Moreover, the incorporation of the abundance regularizer contributes to the improvement of unmixing accuracy. EMLM-Net (DR) outperforms EMLM-Net (D), showing an approximately 3% increase in RMSE and MSAD. In EMLM-Net, considering the regularizers for both abundances and nonlinear parameters leads to further improvement in unmixing accuracy compared to EMLM-Net (DR), with an increase of about 10% in RMSE and 18% in MSAD. This indicates that effectively leveraging the spatial information in HSIs guides the network to achieve superior unmixing accuracy.

Through ablation analysis, the innovations of this article have been validated to significantly improve unmixing accuracy. The use of a dual-stream network structure exhibits the most notable enhancement. Further refinement is achieved by incorporating regularizers for both abundances and nonlinear parameters.

D. Comparison of EMLM-Net With Other Methods

1) Compared Methods: The proposed method is compared with eight SOTA unmixing methods, including three linear unmixing methods (VCA [4] + FCLS [5], uDAS [27], and ADMMNet [36]) and five nonlinear unmixing methods (EGU-Net [38], DMBU [30], HapkeCNN [40], 3D-CNNAE [42], and BNLSU [22]). Notably, VCA [5] + FCLS [3], a commonly used pairing for endmember extraction and abundance estimation in unmixing, is referred to as V/F for brevity. uDAS [35] is an untied denoising AE with sparsity designed for unsupervised unmixing. ADMMNet [46] is an unsupervised unmixing method using an ADMM-based deep unfolding network. EGU-Net [38], designed without reliance on a specific spectral mixture model, is a self-supervised unmixing network that employs convolutional operations for spatial–spectral unmixing. DMBU [39] employs hierarchical BMM for unsupervised unmixing. HapkeCNN [40] is an unsupervised unmixing method for intimate mixtures using the Hapke model and CNN. Based on a general mixture model consisting of a linear mixture part and an additive nonlinear mixture part, 3D-CNNAE [42] employs the 3D-CNN AE network for unsupervised unmixing. Finally, BNLSU [22] is supervised unmixing method using the EMLM, which estimates abundances with fixed endmembers extracted by the VCA. All parameters for the compared methods were fine-tuned empirically or set according to the recommendations from the original literature.

2) Experiments on the Laboratory-Created Dataset: The RMSE and MSAD values obtained by different methods are shown in Table III. For model-based unmixing methods, compared to V/F, the nonlinear unmixing method BNLSU can achieve more accurate abundance estimates. Among the deep learning-based methods, 3D-CNNAE and uDAS, respectively, obtain the optimal RMSE and MSAD. With respect to the deep unfolding-based methods, the proposed EMLM-Net excels ADMMNet, designed for linear unmixing, in both abundance estimation and endmember extraction. Overall, EMLM-Net outperforms other comparative algorithms, indicating that applying deep unfolding techniques to nonlinear unmixing problems can effectively improve unmixing accuracy. For a visual comparison, the generated abundance maps and endmember spectra curves of different methods are provided in Figs. 9 and 10. It is evident that the results obtained by the proposed method closely align with the reference ground truth.

3) Experiments on Real Datasets: In Table IV, the results obtained by different unmixing methods on the Jasper Ridge and Urban datasets are presented. Clearly, the proposed method outperforms the others, further validating its effectiveness in dealing with unmixing problems in real-world scenarios. As shown in Figs. 11 and 12, the abundance maps produced by the proposed method match the reference ones, vividly illustrating the distribution of different endmembers across the scenes. Given the complexity of the Urban dataset, methods other than EMLM-Net yield unsatisfactory abundance maps. Moreover, in Figs. 13 and 14, the endmember spectra extracted by the proposed method are closest to the reference ones. Furthermore, removing the estimated linear mixing component from the original data allows us to estimate the nonlinear intensities associated with different NMMs-based unmixing methods [22]. Fig. 15 illustrates the nonlinear intensity maps, with our method showing a more reasonable distribution. Notably, the nonlinear components exhibit heightened activity at the boundaries or transition zones of various areas, such as water shores and vegetated regions. In conclusion, our method shows advantages in achieving superior unmixing accuracy, yielding refined abundance maps, extracting precise endmember spectra, and clearly reflecting the spatial distribution of nonlinear intensity.

4) Comparison of Noise Robustness: To assess the robustness of different algorithms to noise, we introduce Gaussian noise with SNRs of 15, 20, and 25 dB to the Jasper Ridge dataset and conduct noise robustness experiments based on

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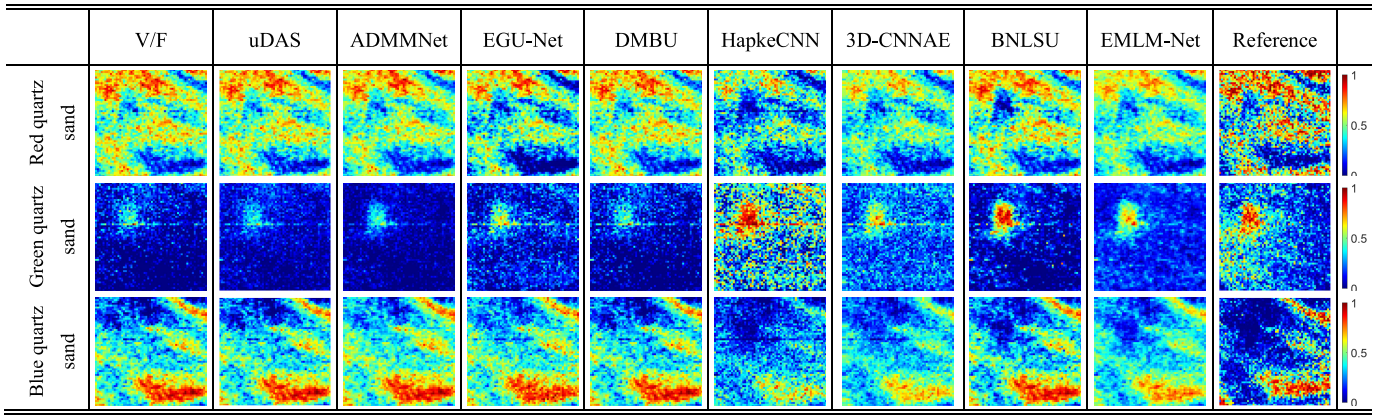
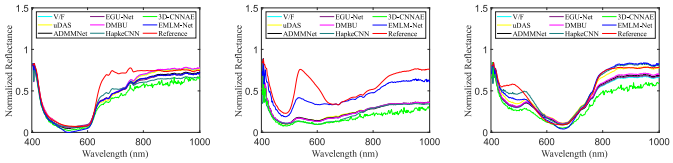


Fig. 9. Abundance maps generated by different methods for the laboratory-created dataset.

TABLE IV
RMSE AND MSAD VALUES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS ON REAL HYPERSPECTRAL DATA

Dataset	Metrics	V/F	uDAS	ADMMNet	EGU-Net	DMBU	HapkeCNN	3D-CNNAE	BNLSU	EMLM-Net
Jasper Ridge	RMSE	0.1430 ±0.0132	0.1419 ±0.0135	0.1262 ±0.0144	0.1256 ±0.0129	0.1642 ±0.0155	0.1394 ±0.0211	0.1543 ±0.0105	0.1132 ±0.0231	**0.0409 ±0.0052**
	MSAD	0.1086 ±0.0134	0.1256 ±0.0207	0.1228 ±0.0208	0.1103 ±0.0110	0.0996 ±0.0068	0.0976 ±0.0139	0.1526 ±0.0097	—	**0.0369 ±0.0044**
Urban	RMSE	0.3408 ±0.0212	0.3291 ±0.0206	0.2610 ±0.0153	0.2665 ±0.0524	0.3481 ±0.0172	0.3214 ±0.0090	0.2860 ±0.0112	0.2779 ±0.0125	**0.1349 ±0.0250**
	MSAD	0.2395 ±0.0040	0.3387 ±0.1116	0.4576 ±0.0845	0.2194 ±0.0124	0.2083 ±0.0179	0.2142 ±0.0051	0.4114 ±0.0304	—	**0.1149 ±0.0211**



(a) (b) (c)

Fig. 10. Endmembers extracted by different methods for the laboratory-created dataset. (a) Red quartz sand. (b) Green quartz sand. (c) Blue quartz sand.

the modified datasets. In Table V, the experimental results obtained by various algorithms at different SNRs are presented.

As SNR decreases, the unmixing accuracy of all methods experiences a decline. Compared to the model- and deep-learning-based unmixing methods, the deep-unfolding-based unmixing methods show superior robustness to noise. In all cases, EMLM-Net produces the best unmixing results. These results emphasize that EMLM-Net not only achieves high unmixing accuracy but also exhibits robustness to noise.

5) Comparison of Run Time: In practical applications, especially those requiring real-time results, the primary concern lies in the speed at which algorithms obtain unmixing results, rather than the speed during the model training phase. Thus, this article only compares the test time of different methods.

Taking the Jasper Ridge dataset as an example, the run time of different methods performing unmixing is compared in Table VI. Clearly, the deep-unfolding-based methods achieve faster unmixing in comparison to model- and deep-learning-based counterparts. Among them, ADMMNet stands out as the fastest, attributed to its simple LMM-based unfolding network structure. Besides, EMLM-Net requires less testing time than the majority of learning-based unmixing methods. Based on the EMLM, BNLSU requires many iterations, while EMLM-Net only requires one forward computation in the prediction/unmixing phase [46]. Therefore, EMLM-Net achieves faster unmixing compared to BNLSU.

E. Discussion

1) High Unmixing Performance: By unfolding the ADMM for the EMLM-based unmixing problem, we construct a network as an encoder and connect it to a decoder derived from the EMLM, creating an AE-like neural network architecture. To leverage the spatial information in HSIs, we do not employ operations like 3D-CNN, considering their possible challenges, such as the high computational complexity and susceptibility to overfitting. Instead, a weight-sharing dual-stream unmixing network is constituted, where both branch networks with the same AE-like architecture perform single data processing. The inputs to the different branch networks are the original HSIs and coarse HSIs. Their outputs, including the estimates of abundances and nonlinear parameters, are utilized to construct regularizers for the loss function. In this way, the local spatial similarity of HSIs can be integrated into the network. Through experiments conducted on laboratory-created and real-world datasets, our method shows superior unmixing accuracy compared to SOTA methods. On the other hand, EMLM-Net shows competitive unmixing efficiency. Unlike model-based unmixing methods that require multiple iterations, EMLM-Net

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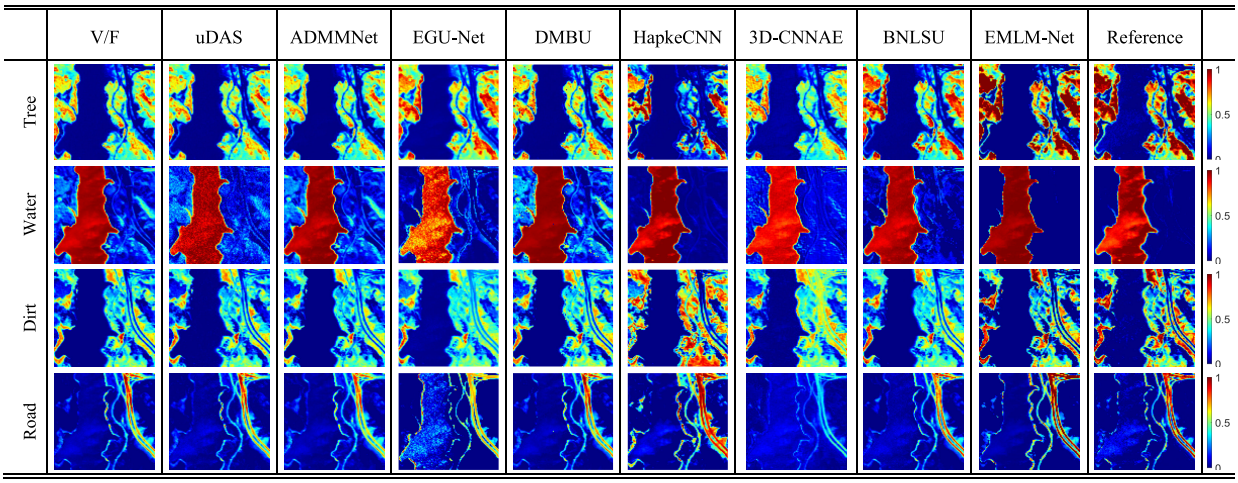


Fig. 11. Abundance maps generated by different methods for the Jasper Ridge dataset.

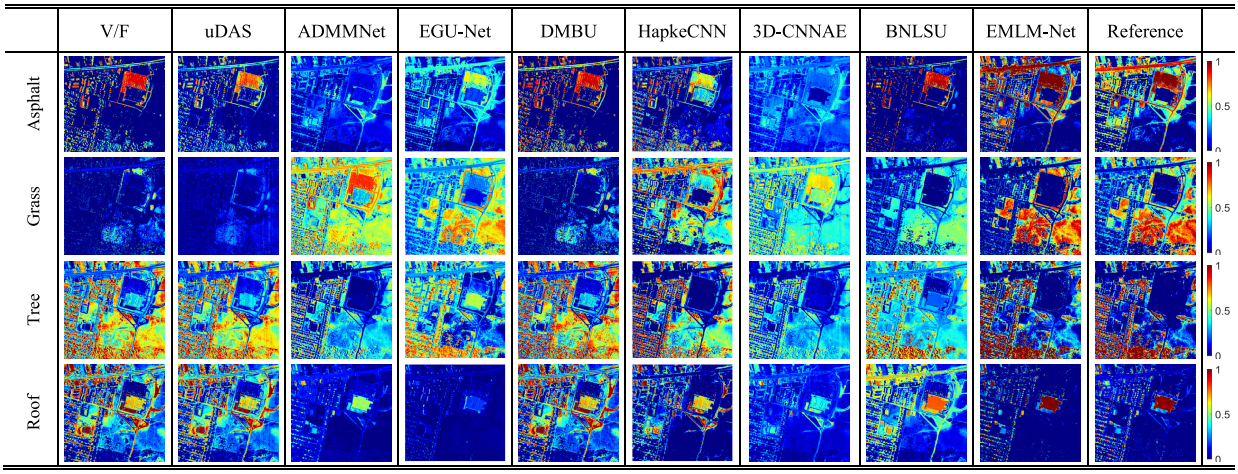


Fig. 12. Abundance maps generated by different methods for the Urban dataset.

TABLE V

RMSE AND MSAD VALUES (MEAN ± STD) OBTAINED BY DIFFERENT METHODS ON THE JASPER RIDGE DATASET WITH VARYING SNRS

SNR	Metrics	V/F	uDAS	ADMMNet	EGU-Net	DMBU	HapkeCNN	3D-CNNAE	BNLSU	EMLM-Net
15 dB	RMSE	0.1610 ±0.0047	0.1775 ±0.0047	0.1376 ±0.0033	0.1755 ±0.0157	0.1782 ±0.0107	0.1472 ±0.0219	0.1823 ±0.0085	0.1645 ±0.0062	**0.0465 ±0.0024**
	MSAD	0.1390 ±0.0097	0.1487 ±0.0151	0.1467 ±0.0068	0.1252 ±0.0158	0.1477 ±0.0104	0.1350 ±0.0109	0.1747 ±0.0094	—	**0.0848 ±0.0049**
20 dB	RMSE	0.1599 ±0.0016	0.1677 ±0.0012	0.1379 ±0.0022	0.1660 ±0.0134	0.1749 ±0.0118	0.1418 ±0.0165	0.1799 ±0.0057	0.1450 ±0.0057	**0.0459 ±0.0070**
	MSAD	0.1181 ±0.0054	0.1437 ±0.0137	0.1209 ±0.0053	0.1226 ±0.0140	0.1155 ±0.0057	0.1124 ±0.0108	0.1610 ±0.0080	—	**0.0496 ±0.0057**
25 dB	RMSE	0.1600 ±0.0011	0.1634 ±0.0018	0.1391 ±0.0012	0.1383 ±0.0082	0.1810 ±0.0161	0.1400 ±0.0089	0.1705 ±0.0078	0.1406 ±0.0029	**0.0430 ±0.0047**
	MSAD	0.1149 ±0.0063	0.1353 ±0.0207	0.1160 ±0.0115	0.1199 ±0.0108	0.1123 ±0.0078	0.1009 ±0.0101	0.1592 ±0.0089	—	**0.0439 ±0.0046**

TABLE VI

COMPARISON OF RUNTIMES FOR DIFFERENT METHODS

Metric	V/F	uDAS	ADMMNet	EGU-Net	DMBU	HapkeCNN	3D-CNNAE	BNLSU	EMLM-Net
Time (s)	1.45	6.13	**0.60**	3.65	14.89	1.85	7.29	160.31	2.81

only involves a single forward computation, enabling faster unmixing. By combining the advantages of both model- and deep-learning-based methods, the proposed method achieves promising unmixing performance.

2) Robustness to Noise: EMLM-Net utilizes weight sharing to benefit from accurate region-based endmember extraction and introduces regularizers in the loss function to constrain abundances and nonlinear parameters. This combination

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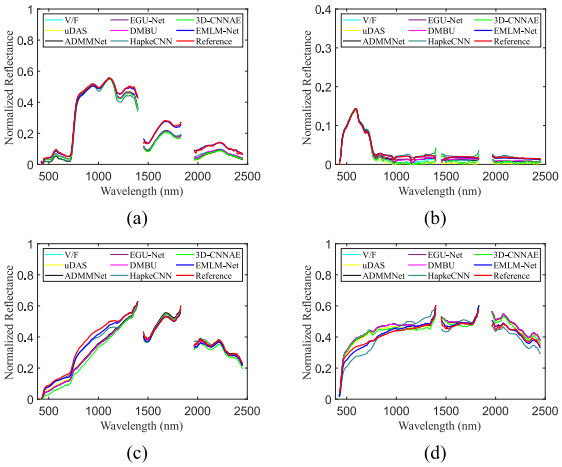


Fig. 13. Endmembers extracted by different methods for the Jasper Ridge dataset. (a) Tree. (b) Water. (c) Dirt. (d) Road.

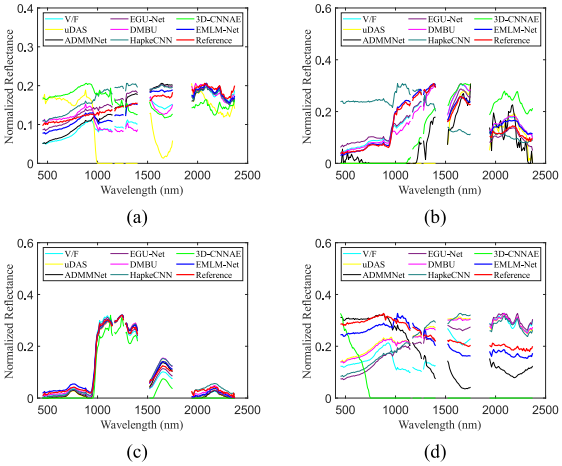


Fig. 14. Endmembers extracted by different methods for the Urban dataset. (a) Asphalt. (b) Grass. (c) Tree. (d) Roof.

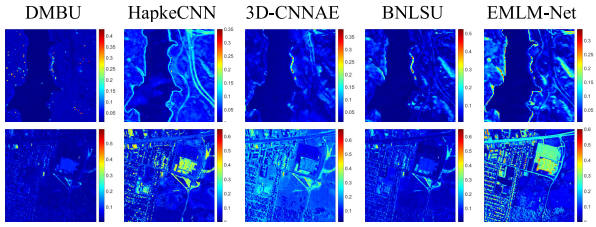


Fig. 15. Nonlinear intensity maps estimated by different methods for the Jasper Ridge dataset (top) and Urban dataset (bottom).

reduces the method's sensitivity to noise and outliers, enhancing its robustness and accuracy in unmixing. Experimental results demonstrate that, in comparison to traditional model- and deep-learning-based unmixing methods, EMLM-Net not only exhibits superior accuracy but also robustness to noise. This highlights the practical applicability of our method, as it maintains good accuracy even in the conditions of low data quality.

3) Strong Interpretability: Compared to typical neural networks, the proposed EMLM-Net shows strong physical interpretability. Each layer of its encoder is designed to align with computational steps in the ADMM algorithm, ensuring explicit functionality and explanation for every component of the network. Consequently, EMLM-Net can readily integrate task-specific prior knowledge, offering trustworthy unmixing results. Furthermore, adopting the interpretable network structure and utilizing the prior information enhance the proposed network's generalization ability.

4) Good Scalability: Embedded within a general dual-stream network architecture facilitating coarse-to-fine learning of spatial features in HSIs and enhanced by local spatial similarity regularizers, the proposed deep unfolding-based AE-like network can be easily adjusted to accommodate real scenarios' nonlinear mixing patterns and chosen iterative solvers. This highlights the good scalability of the proposed method.

V. Conclusion

Existing model- and deep-learning-based nonlinear unmixing methods have limitations, such as complex computational process and poor interpretability. Motivated by the benefits of deep unfolding techniques applied to linear unmixing, we proposed EMLM-Net, an EMLM-inspired dual-stream network, for unsupervised nonlinear HU in this article. EMLM-Net is a weight-sharing dual-stream unmixing network designed with identical AE-like network architectures in both branches to facilitate unsupervised training. The encoder part of this architecture is constructed using a deep unfolding network, with each layer corresponding to one iteration of the ADMM algorithm for the EMLM-based constrained optimization problem, and the decoder part is derived from the EMLM. Moreover, the two branch networks of EMLM-Net take the original HSIs and the corresponding coarse HSIs as inputs, respectively. Their output estimates of abundances and nonlinear parameters are utilized to construct local spatial similarity regularizers for the loss function. In this manner, the spatial information in HSIs can be effectively leveraged, guiding the network toward accurate unmixing results. Experimental results on the laboratory-created dataset and real-world datasets have demonstrated that our method has significant advantages in nonlinear unmixing over the SOTA methods. Combining the advantages of model- and deep-learning-based methods, our proposed EMLM-Net achieves high unmixing performance, robustness to noise, strong interpretability, and good scalability.

In the future work, we intend to improve unmixing accuracy by combining the nonlocal spatial similarity prior of HSIs with deep unfolding networks. Besides, while our method achieves good unmixing results on real datasets containing the spectral variability, it does not specifically address this issue in its design. Exploring the integration of deep unfolding techniques to construct unmixing networks that account for spectral variability remains a promising topic for the future research.

Appendix

Specific computation steps in Algorithm 1 are as follows.

To optimize $\boldsymbol{a}^{k+1}$ at the $(k+1)$th iteration, terms without $\boldsymbol{a}$ are ignored, leading to the simplified optimization subproblem in (24), for which the closed-form solution is given in (25)

$$\boldsymbol{a}^{k+1} = \arg\min_{\boldsymbol{a}} \frac{\mu}{2}\left(\left\|\boldsymbol{E}\boldsymbol{a}^{k} - \boldsymbol{v}_1^{k} - \boldsymbol{d}_1^{k}\right\|_2^2 + \left\|\boldsymbol{a}^{k} - \boldsymbol{v}_2^{k} - \boldsymbol{d}_2^{k}\right\|_2^2\right) \tag{24}$$

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$$\boldsymbol{a}^{k+1} = \left(\boldsymbol{E}^\top \boldsymbol{E} + \boldsymbol{I}_{R\times R}\right)^{-1}\left(\boldsymbol{E}^\top\left(\boldsymbol{v}_1^k + \boldsymbol{d}_1^k\right) + \left(\boldsymbol{v}_2^k + \boldsymbol{d}_2^k\right)\right). \quad (25)$$

The subproblem with respect to $\boldsymbol{p}$, along with its solution, is provided in the following:

$$\boldsymbol{p}^{k+1} = \arg\min_{\boldsymbol{p}} \frac{1}{2}\left\|\boldsymbol{x} - (\boldsymbol{1}_N - \boldsymbol{p}) \odot \boldsymbol{v}_1^k - \boldsymbol{p} \odot \boldsymbol{v}_1^k \odot \boldsymbol{x}\right\|_2^2 + \frac{\mu}{2}\left\|\boldsymbol{p} - \boldsymbol{v}_3^k - \boldsymbol{d}_3^k\right\|_2^2 \quad (26)$$

$$\boldsymbol{p}^{k+1} = \frac{\left(\boldsymbol{x} - \boldsymbol{v}_1^k\right) \odot \left(\boldsymbol{v}_1^k \odot \boldsymbol{x} - \boldsymbol{v}_1^k\right) + \mu\left(\boldsymbol{v}_3^k + \boldsymbol{d}_3^k\right)}{\left(\boldsymbol{v}_1^k \odot \boldsymbol{x} - \boldsymbol{v}_1^k\right)^2 + \mu}. \quad (27)$$

In the $(k+1)$th iteration, the optimization of the subproblem for $\boldsymbol{v}_1$ is expressed in (28), with the solution presented in (29)

$$\boldsymbol{v}_1^{k+1} = \arg\min_{\boldsymbol{v}_1} \frac{1}{2}\left\|\boldsymbol{x} - \left(\boldsymbol{1}_N - \boldsymbol{p}^{k+1}\right) \odot \boldsymbol{v}_1 - \boldsymbol{p}^{k+1} \odot \boldsymbol{v}_1 \odot \boldsymbol{x}\right\|_2^2 + \frac{\mu}{2}\left\|\boldsymbol{E}\boldsymbol{a}^{k+1} - \boldsymbol{v}_1 - \boldsymbol{d}_1^k\right\|_2^2 \quad (28)$$

$$\boldsymbol{v}_1^{k+1} = \frac{1/\mu\left(\boldsymbol{1}_N - \boldsymbol{p}^{k+1} + \boldsymbol{p}^{k+1} \odot \boldsymbol{x}\right) \odot \boldsymbol{x} + \boldsymbol{E}\boldsymbol{a}^{k+1} - \boldsymbol{d}_1^k}{1/\mu\left(\boldsymbol{1}_N - \boldsymbol{p}^{k+1} + \boldsymbol{p}^{k+1} \odot \boldsymbol{x}\right)^2 + \boldsymbol{1}_N}. \quad (29)$$

The variable $\boldsymbol{v}_2$ is then optimized to induce sparsity in the abundances. The subproblem concerning $\boldsymbol{v}_2$ is formulated in (30), and the solution for $\boldsymbol{v}_2^{k+1}$ in (31) is obtained by applying the soft thresholding operator

$$\boldsymbol{v}_2^{k+1} = \arg\min_{\boldsymbol{v}_2} \lambda\left\|\boldsymbol{v}_2\right\|_1 + \frac{\mu}{2}\left\|\boldsymbol{a}^{k+1} - \boldsymbol{v}_2 - \boldsymbol{d}_2^{k+1}\right\|_2^2 + \tau_B(\boldsymbol{v}_2) \quad (30)$$

$$\boldsymbol{v}_2^{k+1} = \max\left(soft\left(\boldsymbol{a}^{k+1} - \boldsymbol{d}_2^k, \lambda/\mu\right), 0\right). \quad (31)$$

On the other hand, regarding the variable $\boldsymbol{v}_3$ associated with the smoothness regularizer, (6) simplifies to the optimization subproblem (32), the solution of which is provided in (33)

$$\boldsymbol{v}_3^{k+1} = \arg\min_{\boldsymbol{v}_3} \frac{\tau}{2}\boldsymbol{v}_3^\top \boldsymbol{L}\boldsymbol{v}_3 + \frac{\mu}{2}\left\|\boldsymbol{p}^{k+1} - \boldsymbol{v}_3 - \boldsymbol{d}_3^k\right\|_2^2 + \tau_C(\boldsymbol{v}_3) \quad (32)$$

$$\boldsymbol{v}_3^{k+1} = \min\left(\left(\tau\boldsymbol{L} + \mu\boldsymbol{I}_{N\times N}\right)^{-1}\left(\mu\left(\boldsymbol{p}^{k+1} - \boldsymbol{d}_3^k\right)\right), \boldsymbol{1}_N\right). \quad (33)$$

Finally, the dual variables $\boldsymbol{d}_1$, $\boldsymbol{d}_2$, and $\boldsymbol{d}_3$ are updated as

$$\boldsymbol{d}_1^{k+1} = \boldsymbol{d}_1^k - \left(\boldsymbol{E}\boldsymbol{a}^{k+1} - \boldsymbol{v}_1^{k+1}\right) \quad (34)$$

$$\boldsymbol{d}_2^{k+1} = \boldsymbol{d}_2^k - \left(\boldsymbol{a}^{k+1} - \boldsymbol{v}_2^{k+1}\right) \quad (35)$$

$$\boldsymbol{d}_3^{k+1} = \boldsymbol{d}_3^k - \left(\boldsymbol{p}^{k+1} - \boldsymbol{v}_3^{k+1}\right). \quad (36)$$

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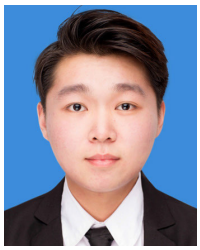
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